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Convergence of shadow economy across U.S. states

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ABSTRACT

Using a novel shadow economy measure constructed from a two-sector dynamic deterministic general equilibrium model, this paper provides the first piece of evidence on the convergence dynamics of the shadow economy at the U.S. state level over the period 1999–2019. The result of the panel club convergence test demonstrates the absence of absolute convergence but also identifies the presence of six club memberships without an apparent regional conglomeration. Further analysis based on the ordered logit regression uncovers that factors such as gross state product, the share of higher education, government size, the share of union participation, capital tax rate, institutional quality, and unemployment rate significantly influence the club membership of the shadow economy. Understanding these findings is imperative for developing policies to curtail the spread of shadow economies across the states in the U.S.

KEYWORDS

Shadow economy; club convergence; ordered logit model; U.S. states

JEL CLASSIFICATION

O11; O18; O47; R11

I. Introduction

Shadow economy activities, which represent all market-based economic activities, are operated outside official authorities for monetary, regulatory, and institutional purposes (Medina and Schneider 2019). These activities often undermine governments' efforts to generate taxes and control the economy, being commonly enhanced by people's avoidance of the costs associated with corruption, high taxes, and poor governance (Canh, Schinckus, and Dinh Thanh 2020). Both researchers and policy-makers alike have been interested in better understanding the nature of the shadow economy due to its economic size¹ and the challenge of the cost incurred by an economy. Moreover, this concern has been fuelled by a burgeoning literature devoted to studying the relationship between the shadow economy and its determinants (F. Schneider 2011; F. G. Schneider and Enste 2013), which further emphasizes its significance.

Though shadow economies are influential to a country, little is known about their steady state (equilibrium) and convergence dynamics. For example, evidence suggests that shadow economies

exhibit significant spillovers across national borders (Goel and Saunoris 2014) and U.S. states (Goel and Saunoris 2016). The cross-border mobility of the shadow economies may indicate that they achieve absolute convergence (same steady state or single equilibrium). Alternatively, shadow economies within specific geographic proximity may converge to a similar steady state, leading to multiple equilibria (i.e. convergent clubs). Recently, there have been two studies that address the nature of convergence of the shadow economy *at the national level* (Gutiérrez-Romero 2021; Henriquez Mora and Saunoris 2023). The conclusions of these studies are qualitatively similar in the sense that absolute convergence is rejected while the convergence clubs are identified. To the best of our knowledge, a specific study examining this topic *at the state level* remains unexplored in the literature. Therefore, the current study bridges this research gap by adding depth to the understanding of the convergence issue of the shadow economy at the U.S. state level.

The contributions of this research are threefold. First, this is the first study to identify the club

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¹The average of estimated shadow economies over 157 countries during the period 1991–2017 accounts for 30.9% of GDP (Medina and Schneider 2019).

📎 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/13504851.2024.2332547>

convergence of the shadow economy at the U.S. state level through the panel convergence test² developed by Phillips and Sul (2007). Second, we adopt a new measure of the shadow economy based on the two-sector dynamic deterministic general equilibrium model (Marshall et al. 2023). Third, our study attempts to identify the determinants of club formation of the shadow economy across the U.S. states. The findings are expected to serve as an invaluable tool in developing targeted policies to curtail the spread of shadow activity in the heterogeneous U.S. society.

The remainder of the paper is structured as follows. Section 2 describes the data. Section 3 discusses our findings. Finally, Section 4 concludes.

II. Data

We utilize the annual shadow economy size³ data for the 50 U.S. states measured by Marshall et al. (2023) for the period from 1999 to 2019. This updated measure shows improvements over the previous one (Wiseman 2013) for two reasons. First, it is estimated based on a two-sector deterministic general equilibrium model (Marshall et al. 2023) that exploits the trends in the observed formal economy, which displays more variation and captures reality more closely. Additionally, the data extends beyond the period of the Great Recession,⁴ which serves as a significant duration that allows for the evaluation of the time-varying behavior of the shadow economy.

III. Findings

Table 1 presents the primary results of the club convergence analysis. In Panel A of Table 1, we test for the full panel absolute convergence. Given that the total sample t -statistic for the log t -test is -31.151 , we consequently reject the null hypothesis of absolute convergence (the t -statistic is smaller than the critical value of -1.65 at 5% significant level) and conclude that the shadow economies in

U.S. states do not converge towards a common steady state. Nonetheless, aligning with Phillips and Sul's (2007) argument that states' outcomes can converge in clusters or clubs, we employ the clustering algorithm to examine the existence of convergence clubs for states sharing a similar pace of convergence.

Panel B of Table 1 displays the results obtained from the club-clustering algorithm and the corresponding club memberships. These results highlight the presence of six distinct clubs (after club merging) and two divergent states. Notably, Clubs 6 and 4 exhibit the fastest and second fastest convergence speeds respectively, as indicated by their large positive beta coefficients (2.107 and 0.561). Moreover, the t -statistics for the log t -test for all clubs are positive and exceed the critical value of -1.65 . Consequently, we do not reject the null hypothesis of club convergence. It follows that the evolution of the shadow economy in the U.S. shows significant differences across states.⁵ Summary statistics for each club are reported in Panel C. Notably, Club 1 despite comprising merely four states, represents the largest average size of shadow economy (15%⁶ of states' formal output). Club 3 is the most populous in terms of club membership, embracing 30% of the states. On average, the size of Club 3 is 12% of the state's formal output, slightly less than Club 1's average. Lastly, Club 6's average size constitutes only 6% of the state's formal output.

Figure 1 depicts that a regional conglomeration of club memberships is not readily discernible. While members of the different clubs are dispersed throughout the country, we do observe some indications that states within the same club tend to be geographically proximate, albeit with a few exceptions.

Further analysis is conducted to shed light on the macroeconomic and demographic variables potentially influencing a state's shift in club membership. To achieve this, we rank the six final clubs by their average shadow economy size. Interestingly, even though the clubs do not demonstrate any clear regional association, the clusters of shadow economies within each club indicate that

²A detailed description of the Phillips and Sul (2007) test is provided in the Appendix.

³The size of the shadow economy is expressed as the percentage of formal output.

⁴The estimated shadow economy in Wiseman (2013) covers the period between 1997 and 2008 only.

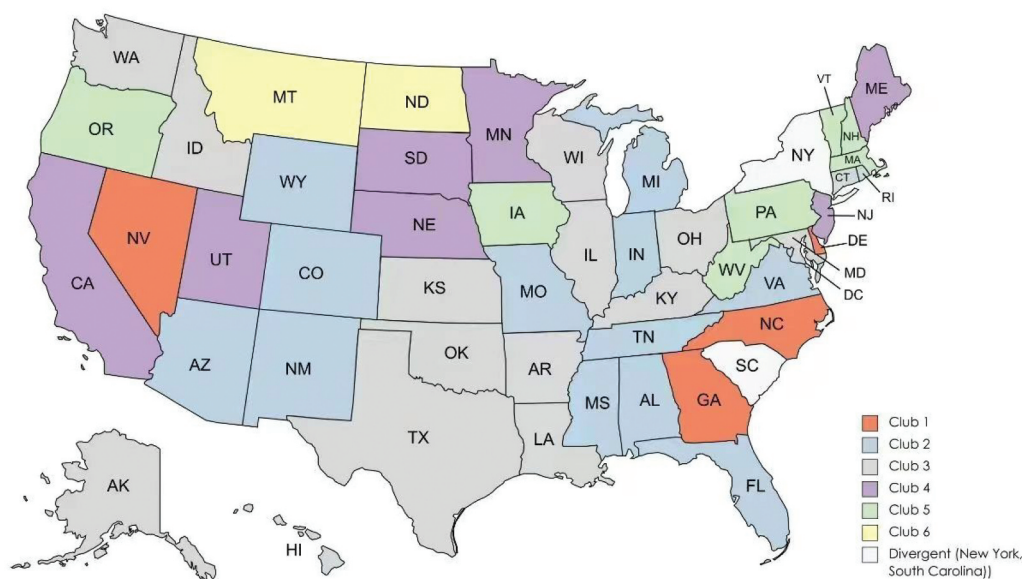
⁵Figures and discussions on the transition paths for different clubs are available upon request.

⁶The mean of different clubs is in Panel C of Table 1 (3rd column).

Table 1. Club convergence results (shadow economy as a percentage of formal output).

Final (Merge club)					
Sample	Beta	Standard error	t-stat	States	
Panel A					
Whole	−1.576	0.041	−31.151	Alabama, Alaska, Arizona, Arkansas, Colorado, Connecticut, Delaware, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Kansas, Kentucky, Louisiana, Maine, Maryland, Michigan, Minnesota, Mississippi, Missouri, Nebraska, Nevada, New Jersey, New Mexico, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, South Carolina, South Dakota, Tennessee, Texas, Virginia, Washington, Wisconsin, Wyoming, Iowa, Massachusetts, New Hampshire, Utah, Vermont, California, Montana, Pennsylvania, Rhode Island, West Virginia, New York	
Panel B					
Club 1 (4)	0.179	0.073	2.442	Delaware, Georgia, Nevada, North Carolina	
Club 2 (12)	0.176	0.155	1.142	Arizona, Florida, Michigan, New Mexico, Alabama, Colorado, Indiana, Mississippi, Missouri, Tennessee, Virginia, Wyoming	
Club 3 (15)	0.168	0.186	0.906	Alaska, Arkansas, Connecticut, Idaho, Illinois, Kansas, Kentucky, Louisiana, Maryland, Ohio, Oklahoma, Oregon, Texas, Washington, Wisconsin	
Club 4 (8)	0.561	0.252	2.224	California, Hawaii, Maine, Minnesota, Nebraska, New Jersey, South Dakota, Utah	
Club 5 (7)	0.195	0.200	0.975	West Virginia, Vermont, Pennsylvania, Rhode Island, New Hampshire, Massachusetts, Iowa	
Club 6 (2)	2.107	0.919	2.293	North Dakota, Montana	
Divergent (2)				New York, South Carolina	
Panel C					
	Number of observation	Mean	Standard deviation	Minimum	Maximum
Club 1	84	0.1514	0.0592	0.0652	0.2458
Club 2	252	0.1316	0.0353	0.0804	0.1894
Club 3	315	0.1227	0.0272	0.0859	0.1792
Club 4	168	0.0967	0.0211	0.0740	0.1446
Club 5	147	0.0982	0.0189	0.0660	0.1325
Club 6	42	0.0661	0.0200	0.0302	0.1093

The figures in the parenthesis denote the number of states.

**Figure 1.** Convergence club classification.

the average shadow economy size (as shown in Figure A1 of the Appendix) is largest in Club 1 (average 15% of GDP). This average size declines down to Club 6 (average 7% of GDP) within the period considered.

Table 2 reports the marginal effects of the ordered logit model at the means, with the final clubs sorted according to their respective shadow economy sizes. The potential determinants of the state-level club membership are drawn from the

Table 2. Ordered logit regression (marginal effects).

Variables	Club 1	Club 2	Club 3	Club 4	Club 5	Club 6
Ln(GDP)	−0.394*** (−0.056)	−0.251*** (−0.0546)	−0.09 (−0.067)	0.1543*** (0.065)	0.2725*** (0.097)	0.309*** (0.023)
Higher education (%)	−0.2473*** (−0.0102)	−0.1813*** (−0.0237)	−0.0006 (−0.0091)	0.0831*** (0.0153)	0.1352*** (0.0147)	0.2179*** (0.0414)
Economic freedom	−0.0062 (−0.0219)	−0.0145 (−0.0552)	−0.0002 (−0.0003)	0.0015 (0.0332)	0.0142 (0.0339)	0.0052 (0.0085)
Union membership (%)	0.041*** (0.00932)	0.1165*** (0.0237)	0.0006 (0.0756)	−0.0643*** (−0.0105)	−0.0543*** (−0.0139)	−0.0388*** (−0.0173)
Capital tax (%)	0.25*** (0.09)	0.179*** (0.0033)	0.002 (0.0027)	−0.151*** (−0.051)	−0.131*** (−0.034)	−0.24*** (−0.05)
Government size	−0.0298*** (−0.0016)	−0.0272*** (−0.0054)	−0.0001 (−0.0014)	0.0131*** (0.0031)	0.0145*** (0.0024)	0.0321*** (0.0009)
Institutional quality	−0.377*** (−0.065)	−0.33*** (−0.053)	−0.05 (−0.059)	0.1541*** (0.052)	0.233*** (0.095)	0.357*** (0.12)
Burden of government regulations	0.08801 (0.0676)	0.06902 (0.0642)	−0.0197 (−0.0523)	−0.02435 (−0.1379)	−0.0464 (−0.0902)	−0.05871 (−0.2047)
Unemployment rate	0.3017*** (0.0842)	0.2043*** (0.0316)	−0.0697 (−0.08)	−0.1053*** (−0.0468)	−0.1164*** (−0.0423)	−0.2346*** (−0.0796)
Average age of retirement	0.0947 (0.0797)	0.1932 (0.128)	−0.01543 (−0.0184)	−0.1035 (−0.0863)	−0.1579 (−0.1026)	−0.0111 (−0.0148)
Observations	1008	1008	1008	1008	1008	1008

This table reports the ordered logit model marginal effects at the means. The dependent variable is the club membership for each state. The ordering of clubs is based on the average size of the shadow economy of each club. A higher club number indicates a lower mean value of the shadow economy. Table A1 of the Appendix provides the definitions of the explanatory variables. Standard errors in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.001$.

literature (Berdiev, Pasquesi-Hill, and Saunoris 2015; Marshall et al. 2023) and incorporate the following variables at the state level: GDP, percentage of higher education, economic freedom, share of union membership, capital tax rate, government size, institutional quality, burden of government regulation, unemployment rate, and average age of retirement.⁷ Table A1 of the Appendix contains detailed variable descriptions in the analysis. Each column in Table 2 represents the estimated marginal effects on the probability of belonging to a club at their respective means of the independent variables. For example, a one percentage point increase in the share of higher education decreases the likelihood of a state belonging to Club 1 (highest average shadow economy) by 24.73% and to Club 2 by 18.13%, while the likelihood of being in Club 6 rises by 20.18%.

The findings in Table 2⁸ suggest that a state's likelihood of joining the higher-ordered club rises if a state displays these characteristics⁹ (in descending order): a higher institutional quality,¹⁰ a higher gross state product, a larger share of higher education, and a larger government size. When all else is equal, more prosperous states increase the

opportunity cost of shadow production and are better equipped to oversee and control shadow activities. As an important demographic factor, a larger proportion of the higher educated population is related to a higher level of human capital. This relationship implies higher wages in the formal sector and effectively increases the opportunity costs of labor participation and consequently, production in the shadow economy. A larger government size also performs an insurance function for the private sector (Dreher and Schneider 2010; Williams 2019), which deters individuals from moving into the shadow economy. A better institutional quality is related to a promotion in resource allocation efficiency (Park 2012). This will provide the opportunity to develop official economic activities. In contrast, the chance of a state being a part of a lower-ordered club increases if a state features these characteristics (in the descending order): A higher unemployment rate, a higher capital tax rate, and a higher share of union membership (which reflects the level of regulatory burdens). Both higher tax rate and higher unionization provide incentives for businesses to manage in the shadow sector due to the increasing costs of

⁷As suggested by the reviewers, we also include institutional quality, burden of government regulation, unemployment rate, and average age of retirement in the analysis.

⁸A graphical representation of Table 2 is exhibited in Figure A2 of the Appendix.

⁹Only significant variables are discussed.

¹⁰The impact of institutional quality on the marginal effect of explanatory variables is presented graphically in figures A3(a) and A3(b) of the Appendix.

operating in the official sector. These results largely corroborate the existing evidence on the macroeconomic and demographic determinants of the shadow economy in U.S. states (Berdiev, Pasquesi-Hill, and Saunoris 2015; Marshall et al. 2023).

IV. Conclusion

Given the significance of the shadow economy in the understanding of economic development, this paper takes an additional step to scrutinize the nature of convergence in the shadow economy across 50 U.S. states between 1999 and 2019. The results reject the absolute convergence and corroborate the existence of six distinct convergence clubs. Analysis of the state-level factors using the ordered logit model reveals that high gross state product, high share of the educated population, large government size, low capital tax rate, low share of union membership, high institutional quality, and low unemployment rate significantly support the maintenance of clubs with low shadow economy. Our findings highlight the importance of accounting for the convergence patterns and their determinants in the shadow economies across states in the U.S. and suggest that states would benefit from a concerted effort in developing policies to mitigate the spread of the shadow economies.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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